Compress Then Explain in Practice: Stable and Generalizable Feature Attributions

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Abstract

Exact calculation of feature attributions is computationally too expensive for continuous model auditing. The Compress Then Explain (CTE) framework accelerates this using optimized background coresets, yet its practical reliability has remained unevaluated. We conduct a rigorous evaluation of CTE against uniform random sampling across 8 models and 30 independent seeds. CTE consistently yields lower approximation errors, reducing global Mean Absolute Error by up to 42.3% at a representative coreset size of $N = 128$. Furthermore, the CTE approximations exhibit lower variance. This provides a stable signal for tracking temporal feature drift and evaluating out-of-sample data, establishing CTE as a robust technique for reliable model maintenance.

1. Introduction

Feature attribution methods like SHAP (Lundberg and Lee, 2017) have become standard for evaluating machine learning models. However, generating exact explanations might be computationally infeasible for large datasets processed by complex models. This bottleneck prevents continuous model monitoring in dynamic environments, where feature importance is exposed to drifts over time. To address this, the Compress Then Explain (CTE) (Baniecki et al., 2025) framework was introduced, aiming to accelerate explanations by using a highly representative coreset, minimizing Maximum Mean Discrepancy (MMD). Yet, its practical fidelity and temporal stability for continuous auditing have lacked appropriate validation.

Contribution. We conduct a rigorous evaluation of CTE against standard uniform random sampling. To isolate the approximation error, we use optimized *TreeExplainer* and *LinearExplainer* to obtain precise ground-truth SHAP values. Our evaluation yields three contributions: **(1) Fidelity advantage:** CTE approximates SHAP values with significantly lower Mean Absolute Error (MAE) for tree and linear models. **(2) Temporal drift tracking:** A fixed CTE background faithfully tracks shifts in feature importance, providing a stable signal compared to random sampling. **(3) Explanation generalization:** CTE backgrounds generated from training data maintain high fidelity when evaluating unseen validation instances, proving their strong generalization.

Related Work. Our study combines dataset summarization and explainable AI. Algorithms like Kernel Thinning (Dwivedi and Mackey, 2024) and Compress++ (Shetty et al., 2022) were originally developed for general distribution compression, providing strict Maximum Mean Discrepancy (MMD) guarantees. Recently, the CTE framework (Baniecki et al., 2025) successfully utilized these general techniques to accelerate feature attribution. While existing literature demonstrates CTE's computational efficiency, we extend it by providing the evaluation of its reliability in dynamic tasks, such as temporal tracking and out-of-sample testing.

2. Methodology and Experimental Setup

Background Compression. Let $X_{train} \in \mathbb{R}^{M \times D}$ denote the full training dataset. We aim to extract a highly representative coreset $X_c \subset X_{train}$ of target size $N \ll M$. To overcome the quadratic complexity of standard thinning algorithms, we use Compress++ (Shetty et al., 2022). This reduces the initial dataset in near-linear time ($\mathcal{O}(M \log^3 M)$), maintaining an MMD bound. Next, we apply Kernel Thinning (Dwivedi and Mackey, 2024) to iteratively halve this intermediate subset until the exact target size N is reached. For MMD calculations, we utilize a Gaussian kernel, setting its bandwidth σ to the median pairwise distance between samples, a standard parameter-free strategy (Gretton et al., 2012).

Dataset and Models. We evaluate our framework on the `ecom_offers` dataset from TabReD (Rubachev et al., 2024), consisting of ∼160k instances and 119 features. We split the data chronologically into training (∼109k), validation (∼24k) and test (∼26k) sets. We train and fine-tune 8 models (XGBoost, Random Forest, LightGBM, CatBoost, Logistic Regression, Ridge, Lasso, SVC) using Optuna (Akiba et al., 2019). Predictive performance details are in Appendix 5.2.

Evaluation Protocol. For each model, we estimate feature attributions on a representative random subset of instances. To establish exact ground truth, SHAP values are computed using the full training set as the background. We approximate these attributions by substituting the full background with CTE coresets and benchmark against a uniform random sampling baseline. Both generation procedures are repeated across 30 independent seeds. We evaluate fidelity using two metrics: Local MAE (average error per instance) and Global MAE (error of the overall feature importance).

3. Experiments and Results

RQ1: Fidelity advantage. Figure 1 shows the Global and Local MAE across different background sizes $N \in \{4, \ldots, 256\}$. CTE yields lower average errors than random sampling for all tested sizes. To illustrate this advantage at a moderate compression level, at $N = 128$, CTE reduces Global MAE by 29.9% for tree models and 42.3% for linear models. In contrast, the slight drop in performance at $N = 256$ reflects an architectural trade-off of Compress++ (Shetty et al., 2022): for our dataset, $N = 256$ is the initial output of the Compress++ algorithm, generated with a fast, near-linear approximation without adjustment, leading to less precise outputs, despite the larger size of the coreset.

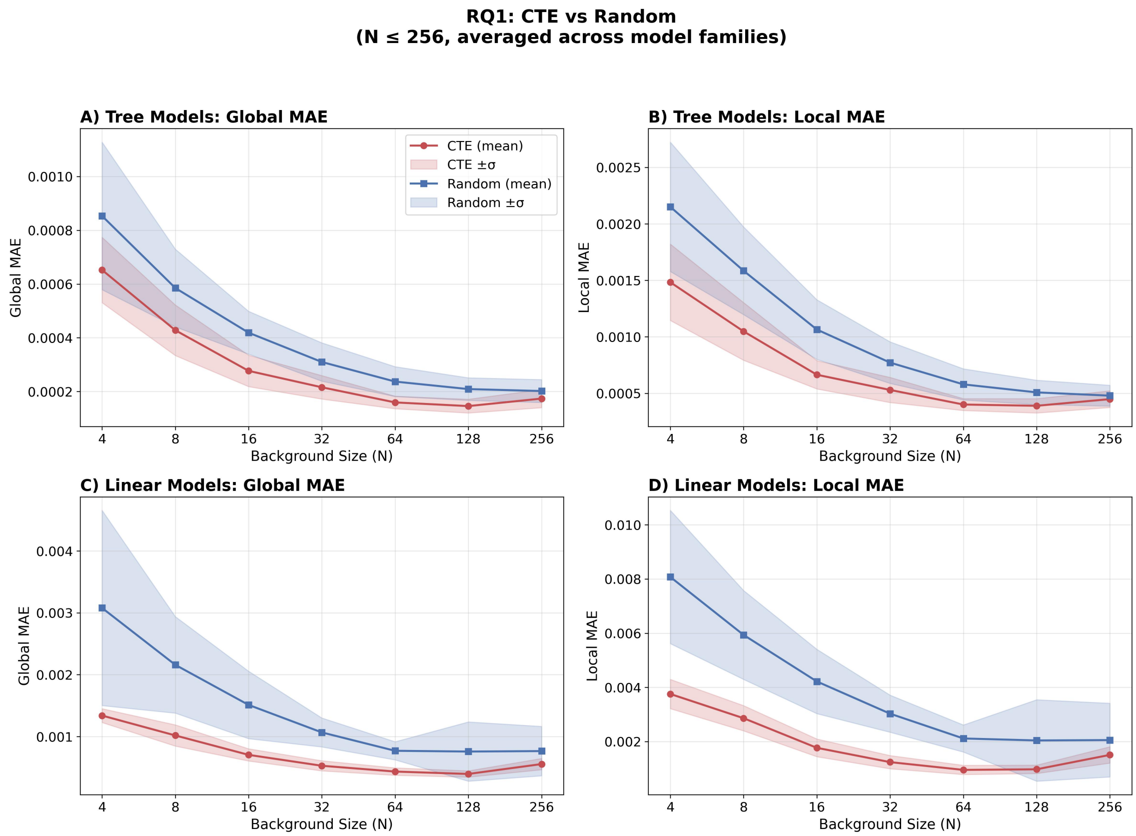

Figure 1: Global and Local MAE of SHAP approximations across varying background sizes. CTE consistently achieves lower error.

RQ2: Temporal drift tracking. In dynamic environments, computing exact SHAP across consecutive windows is computationally expensive. Fixing coreset size to $N = 128$, Figure 2 demonstrates that CTE outperforms random sampling in tracking temporal shifts. Standard sampling introduces high variance, which might lead to misdiagnosing true feature drift. CTE extracts the underlying distribution, providing a stable signal.

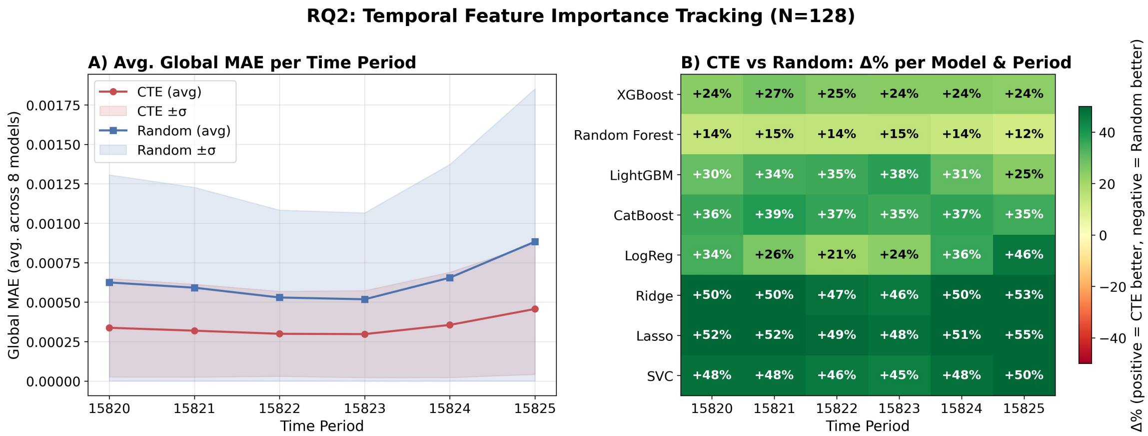

Figure 2: Temporal tracking of feature importance across 5 chronological windows. CTE provides a stable, lower-noise signal.

RQ3: Explanation generalization. Evaluating explanations on unseen data is essential for practical deployments. As detailed in Figure 3 (Appendix 5.1), CTE consistently maintains lower MAE and significantly lower variance than random sampling across both training and validation contexts. Notably, CTE yields comparable or even lower approximation errors on the validation set across the evaluated architectures, indicating robust generalization. The higher variance of random sampling makes it unreliable for precise diagnostics.

4. Conclusion

Our extensive evaluation confirms that the CTE framework is an effective summarization technique that enables continuous model auditing. Beyond accelerating standard attributions, CTE provides a stable signal essential for dynamic monitoring: temporal tracking (RQ2) and explanation generalization (RQ3). It consistently outperforms uniform random sampling in fidelity and reliability.

Limitations and Future Work. Our empirical evaluation is restricted to a single temporal dataset (`ecom_offers`) and models compatible with exact explainers (linear/trees). Future work could test generalizability across deep neural networks and other explanation methods like SAGE (Covert et al., 2020). Additionally, benchmarking CTE against standard sampling using model-agnostic methods like *KernelSHAP* is necessary to further quantify the computational trade-offs of initial distribution compression.

References

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5. Appendix

5.1 Explanation Generalization (RQ3)

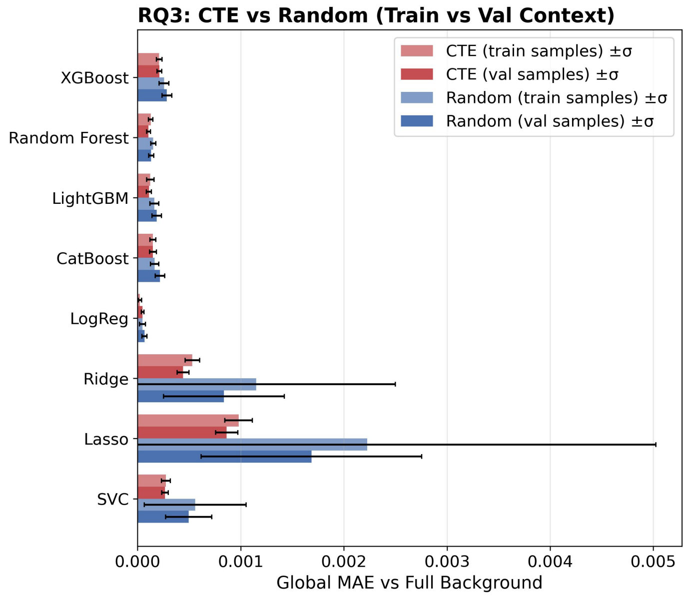

Figure 3: Attribution error on train vs. validation datasets. CTE backgrounds extracted from training data generalize seamlessly to unseen validation data with low variance.

5.2 Predictive Performance of Trained Models

Table 1 reports the performance of the 8 classification models on the `ecom_offers` test set. This dataset is challenging, resulting in modest predictive performance across all architectures. However, the training process shows that the models captured a baseline predictive signal. This learned signal ensures that the evaluated SHAP attributions reflect actual feature dependencies rather than random noise.

5.3 Approximation Stability

Table 2 reports the standard deviation of the Global MAE across 30 independent runs at a background size of $N = 128$. CTE consistently exhibits lower variance than random sampling across all architectures, proving to be a more stable solution for continuous auditing.

Table 1: Predictive Performance of Trained Models on the Test Set

Model	**Family**	**ROC-AUC**	**Accuracy**
XGBoost	Tree	0.5665	0.5996
Random Forest	Tree	0.5680	0.6445
LightGBM	Tree	0.5697	0.6397
CatBoost	Tree	0.5633	0.5961
LogReg	Linear	0.5571	0.5907
Ridge	Linear	0.5745	0.6442
Lasso	Linear	0.5748	0.6458
SVC	Linear	0.5767	0.6272

Table 2: Global MAE Standard Deviation at $N = 128$ (30 independent seeds)

Model	**Family**	**Random Std (σ)**	**CTE Std (σ)**	**Variance Reduction**
XGBoost	Tree	4.85e-05	2.48e-05	**48.9%**
Random Forest	Tree	2.71e-05	1.97e-05	**27.4%**
LightGBM	Tree	4.58e-05	2.43e-05	**46.8%**
CatBoost	Tree	4.66e-05	3.36e-05	**27.9%**
LogReg	Linear	2.45e-05	1.28e-05	**47.8%**
Ridge	Linear	5.93e-04	5.47e-05	**90.8%**
Lasso	Linear	1.07e-03	1.03e-04	**90.4%**
SVC	Linear	2.26e-04	3.24e-05	**85.7%**

5.4 Individual Model Evaluations

The main text reports metrics aggregated across model families. To confirm that these trends hold consistently for individual architectures, this section provides per-model plots evaluating fidelity advantage (RQ1), temporal drift tracking (RQ2) and explanation generalization (RQ3).

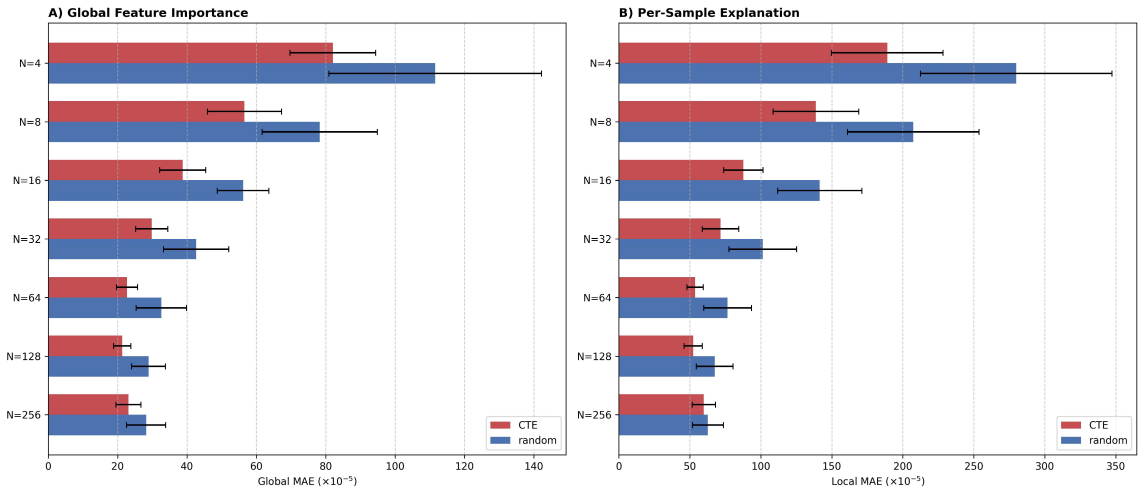

Figure 4: XGBoost: Fidelity Advantage.

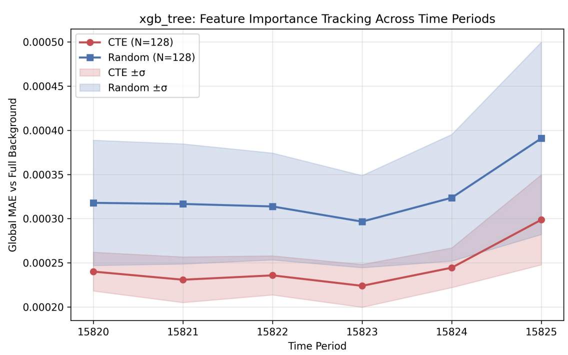

Figure 5: XGBoost: Temporal Drift MAE.

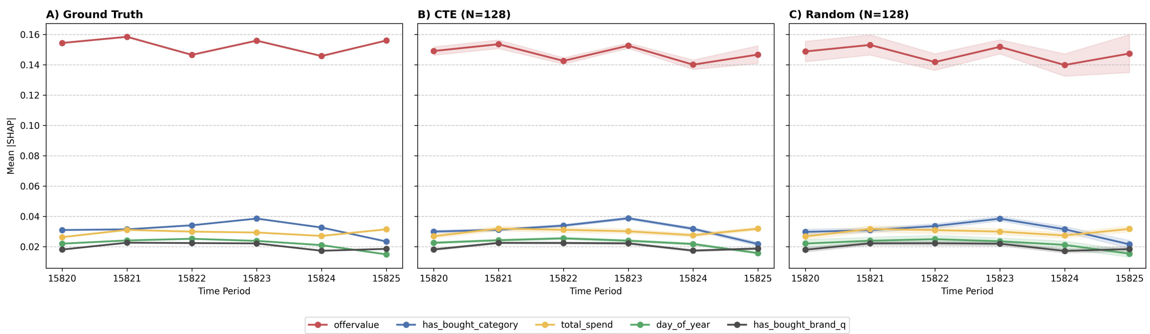

Figure 6: XGBoost: Feature Importance Tracking Over Time.

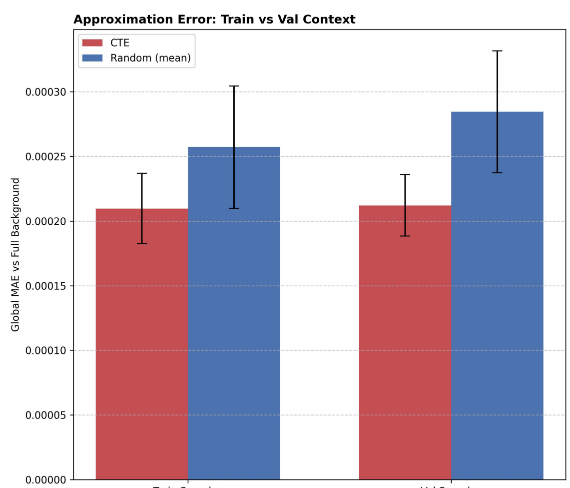

Figure 7: XGBoost: Explanation Generalization.

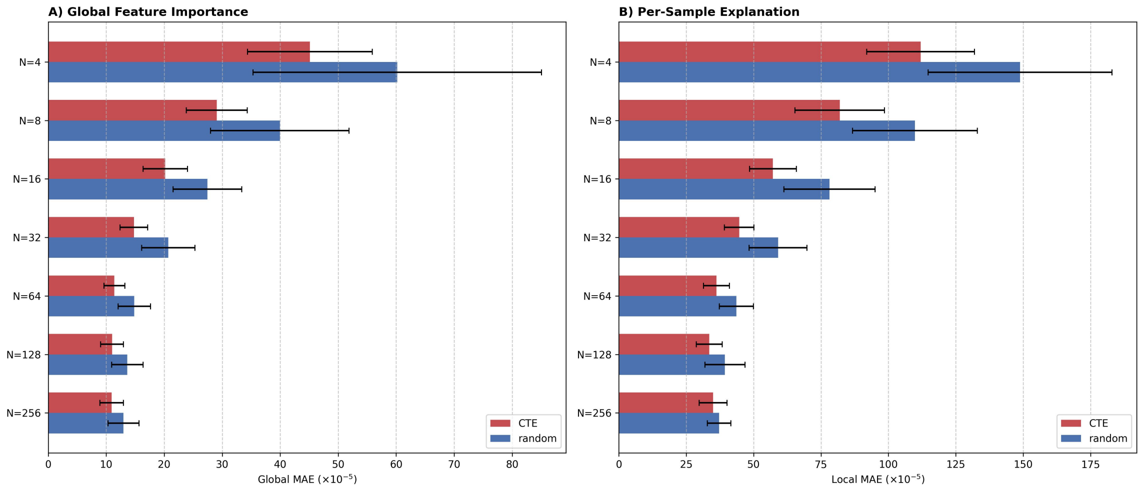

Figure 8: Random Forest: Fidelity Advantage.

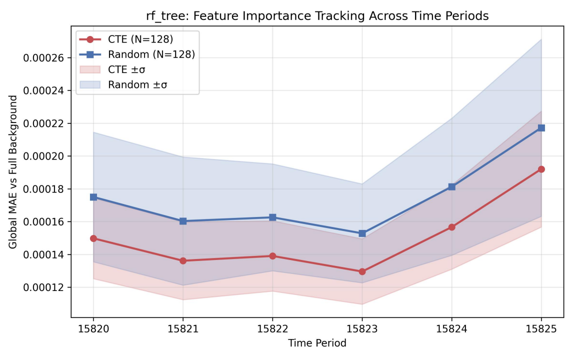

Figure 9: Random Forest: Temporal Drift MAE.

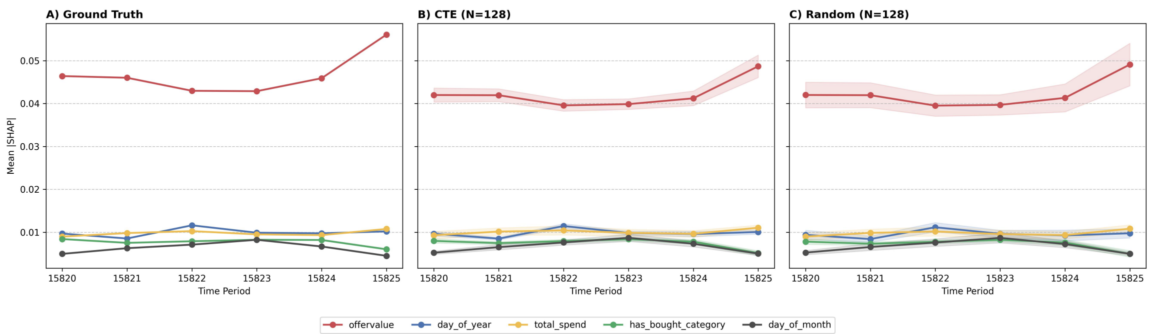

Figure 10: Random Forest: Feature Importance Tracking Over Time.

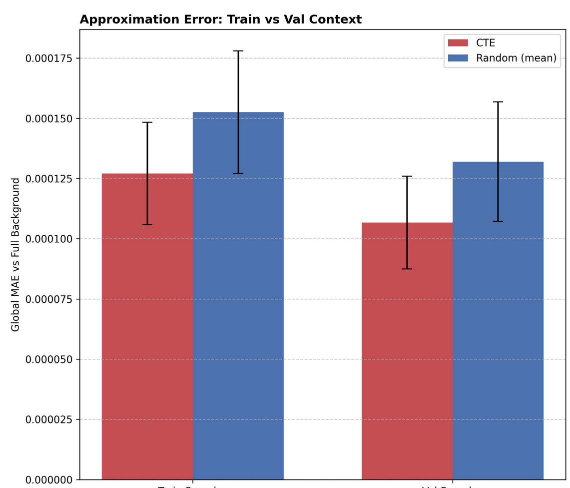

Figure 11: Random Forest: Explanation Generalization.

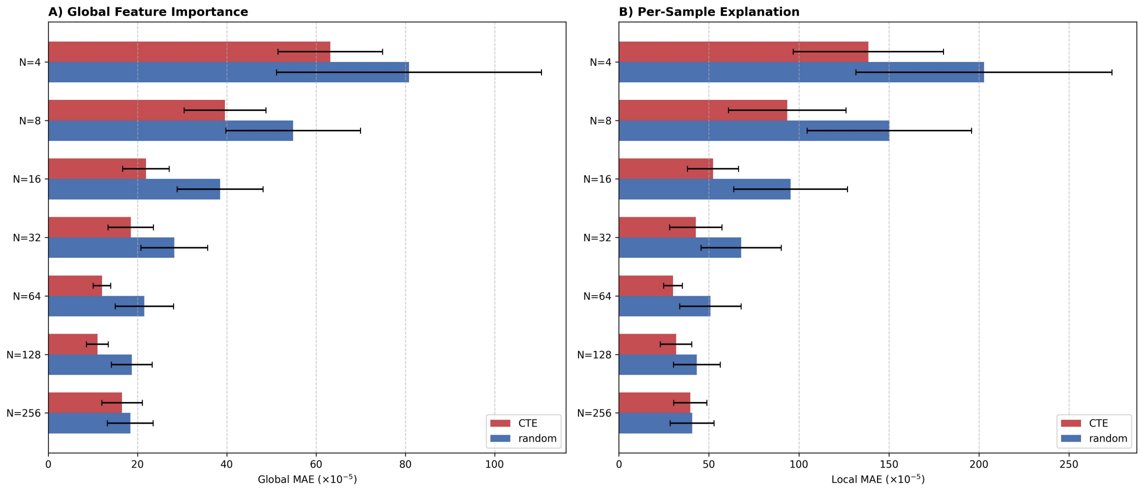

Figure 12: LightGBM: Fidelity Advantage.

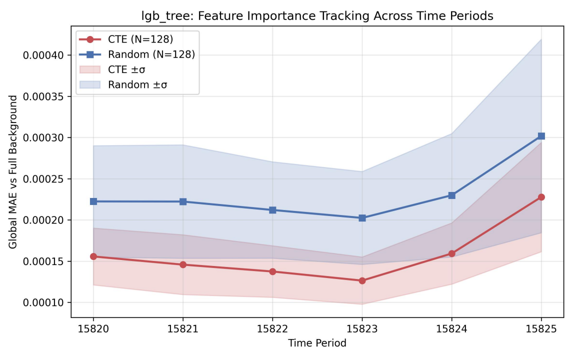

Figure 13: LightGBM: Temporal Drift MAE.

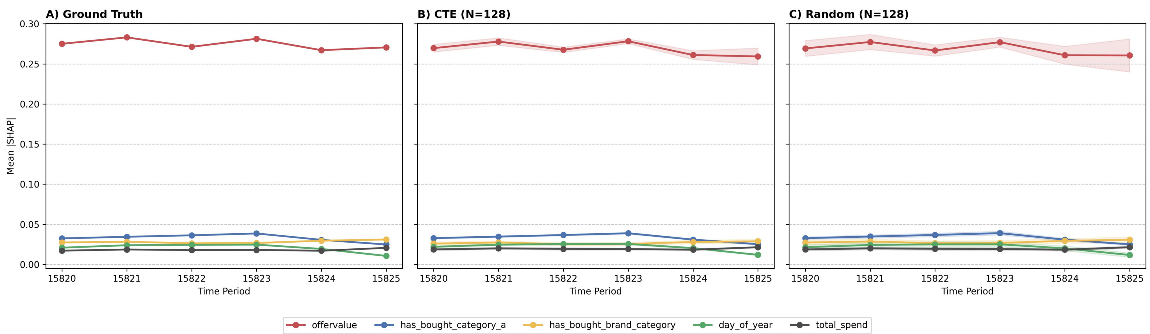

Figure 14: LightGBM: Feature Importance Tracking Over Time.

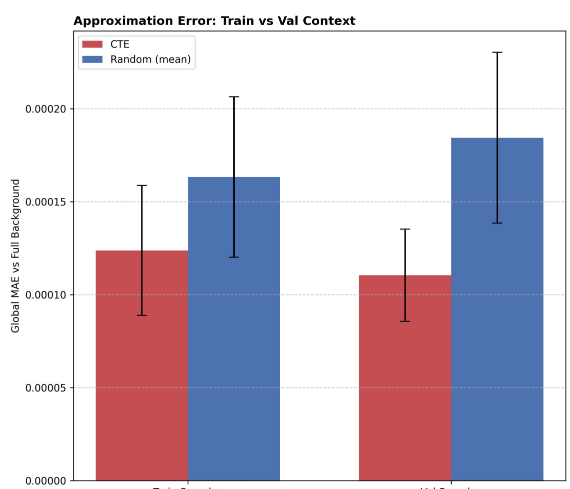

Figure 15: LightGBM: Explanation Generalization.

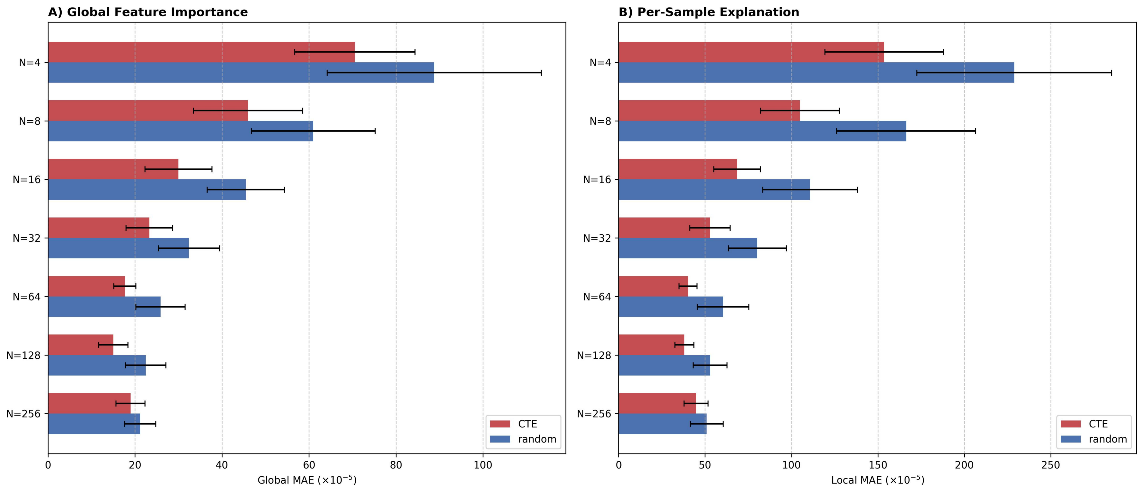

Figure 16: CatBoost: Fidelity Advantage.

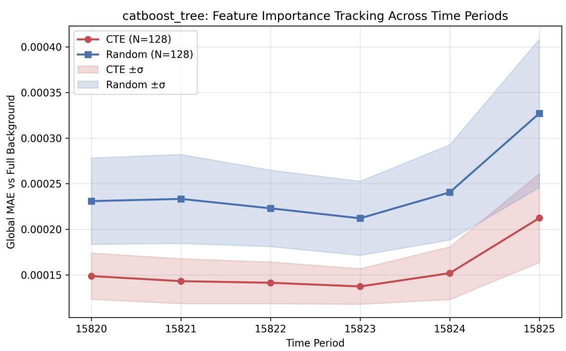

Figure 17: CatBoost: Temporal Drift MAE.

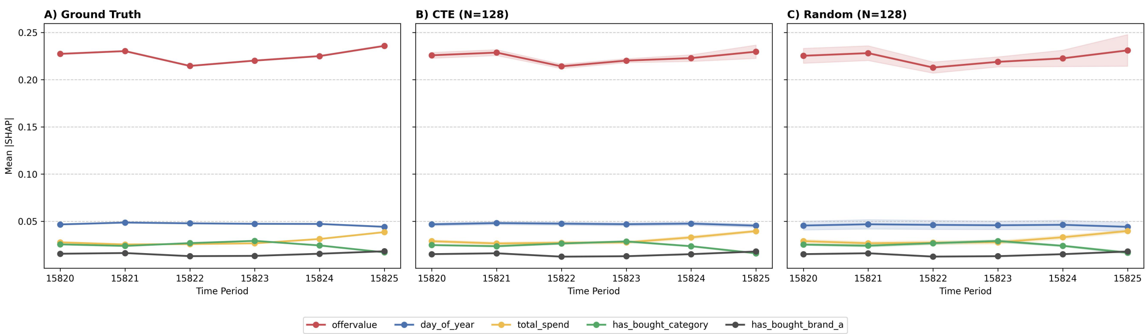

Figure 18: CatBoost: Feature Importance Tracking Over Time.

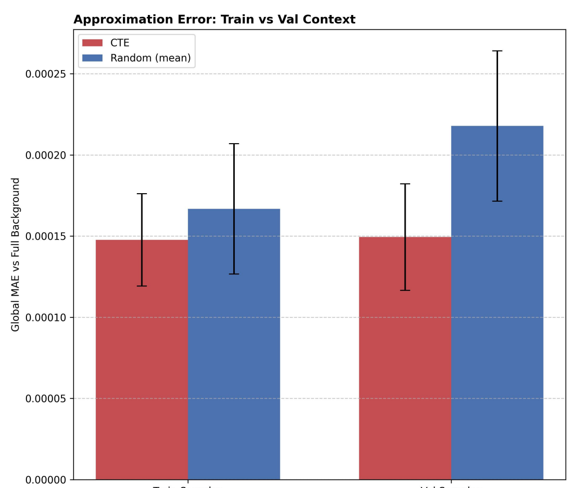

Figure 19: CatBoost: Explanation Generalization.

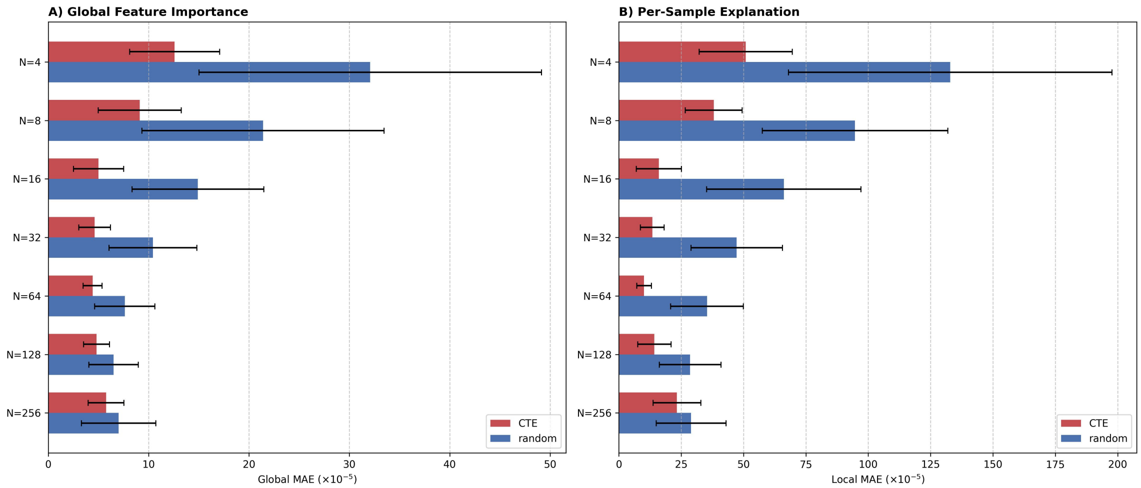

Figure 20: Logistic Regression: Fidelity Advantage.

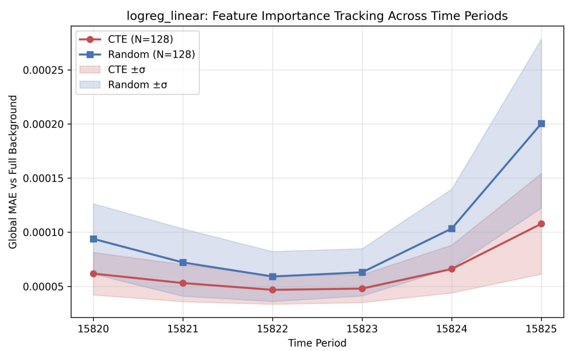

Figure 21: Logistic Regression: Temporal Drift MAE.

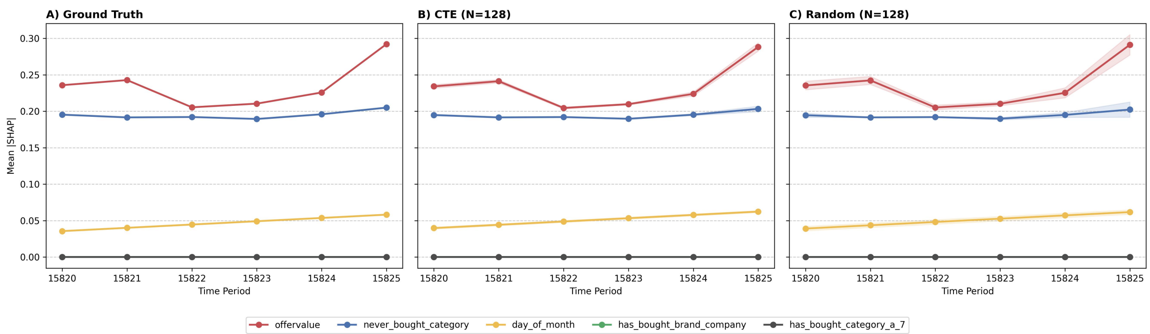

Figure 22: Logistic Regression: Feature Importance Tracking Over Time.

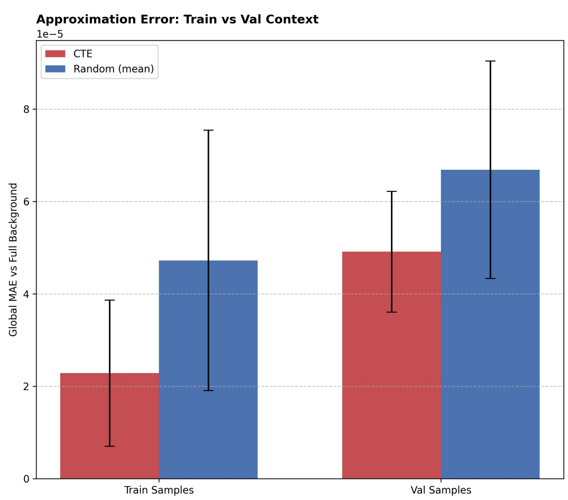

Figure 23: Logistic Regression: Explanation Generalization.

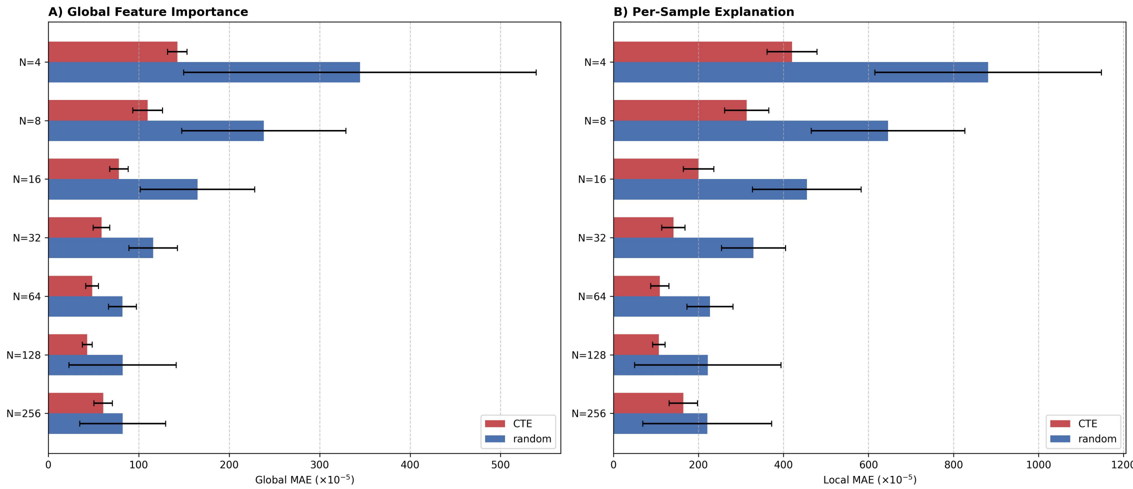

Figure 24: Ridge: Fidelity Advantage.

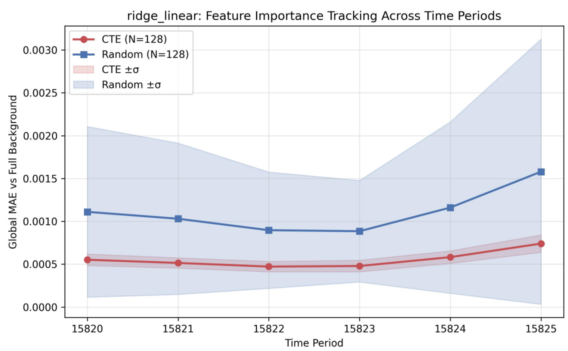

Figure 25: Ridge: Temporal Drift MAE.

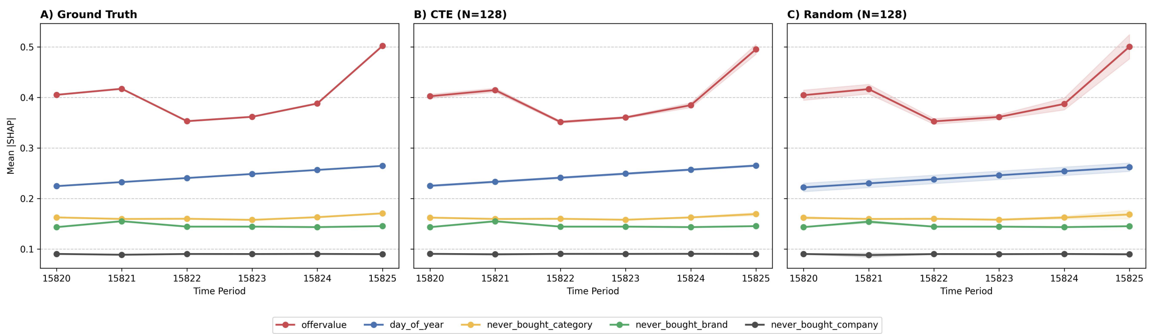

Figure 26: Ridge: Feature Importance Tracking Over Time.

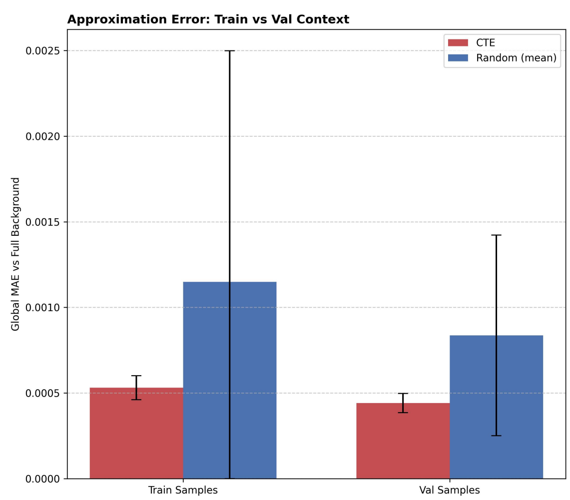

Figure 27: Ridge: Explanation Generalization.

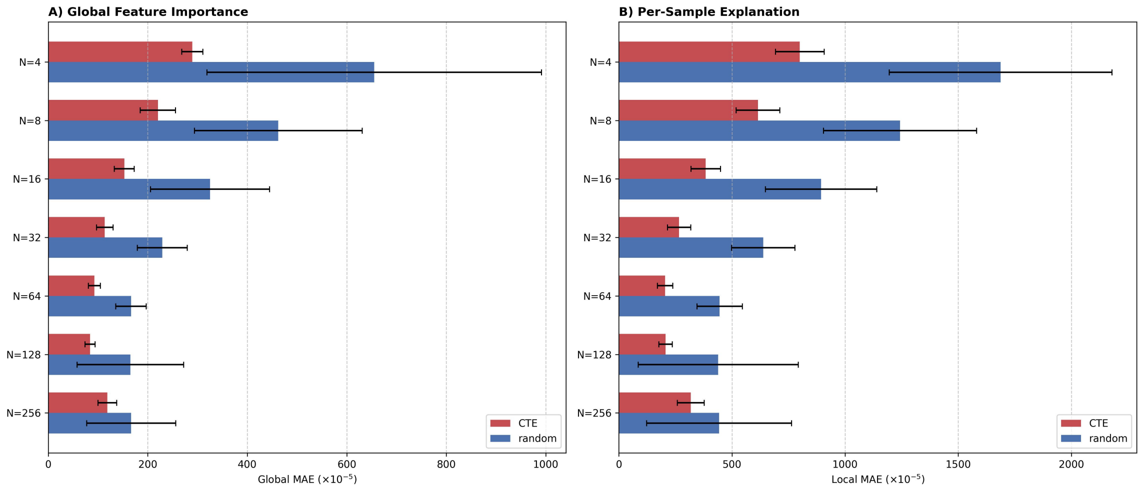

Figure 28: Lasso: Fidelity Advantage.

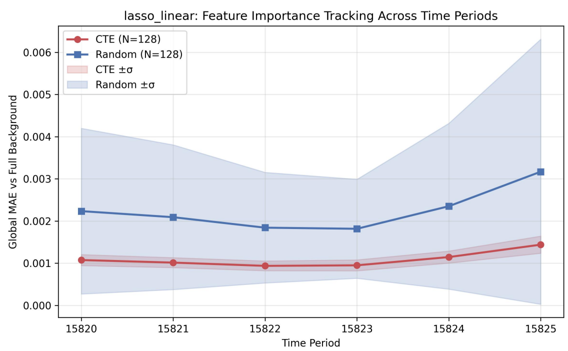

Figure 29: Lasso: Temporal Drift MAE.

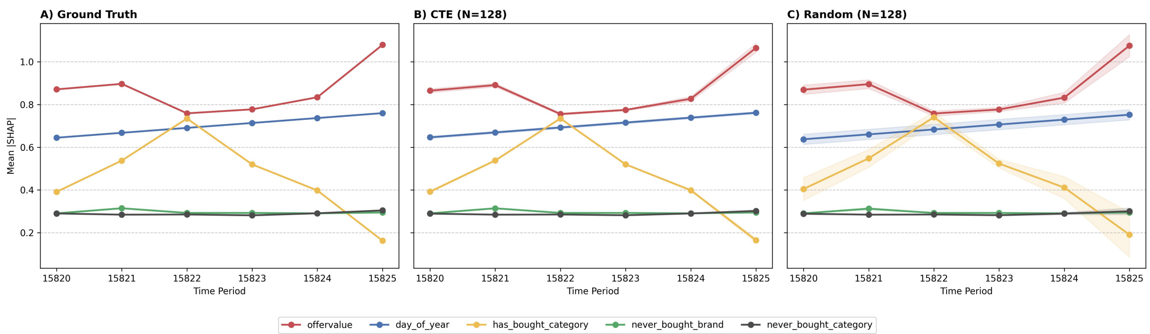

Figure 30: Lasso: Feature Importance Tracking Over Time.

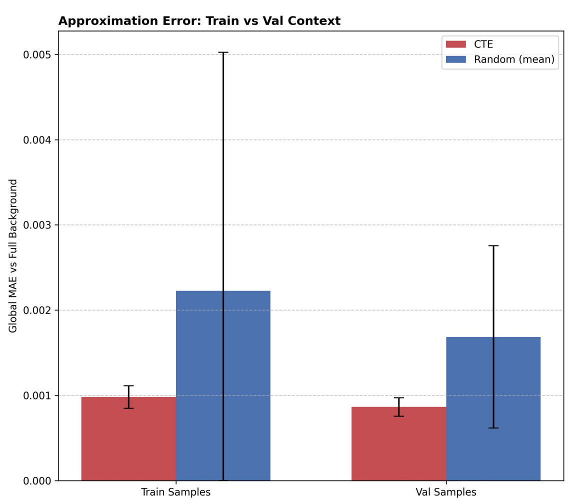

Figure 31: Lasso: Explanation Generalization.

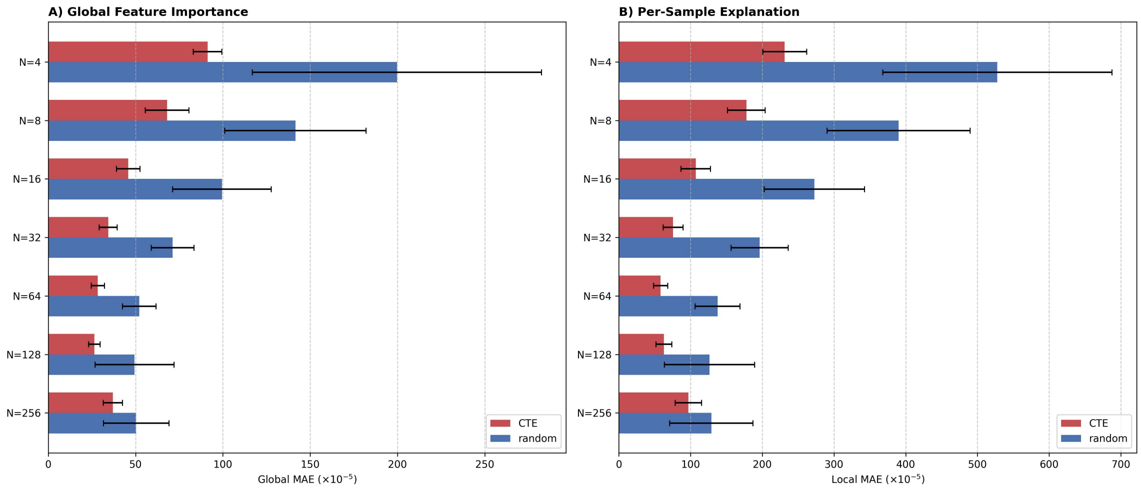

Figure 32: SVC: Fidelity Advantage.

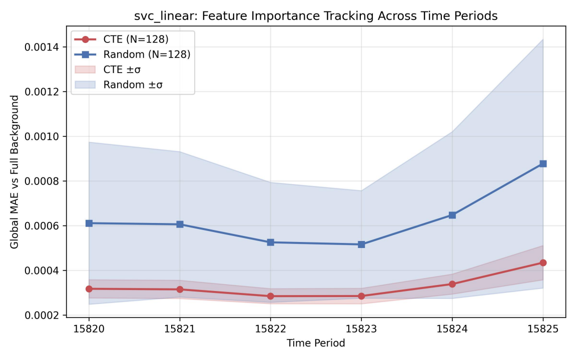

Figure 33: SVC: Temporal Drift MAE.

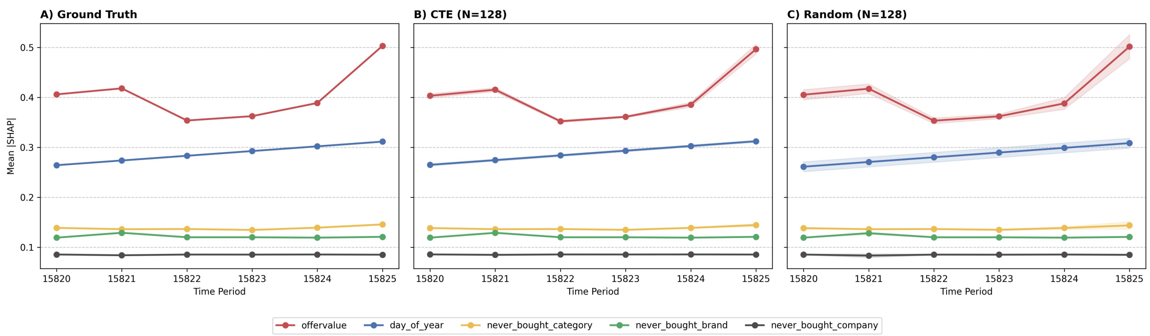

Figure 34: SVC: Feature Importance Tracking Over Time.

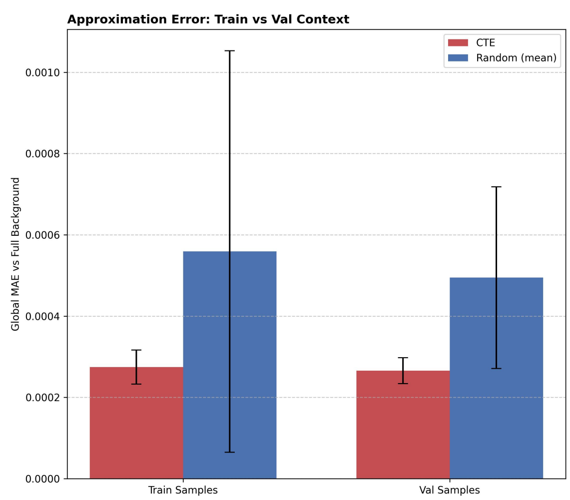

Figure 35: SVC: Explanation Generalization.